



AJKD Atlas of Renal Pathology: Cellular Variant of Focal Segmental Glomerulosclerosis

Agnes B. Fogo, MD,¹ Mark A. Lusco, MD,¹ Behzad Najafian, MD,² and Charles E. Alpers, MD²

Clinical and Pathologic Features

The Columbia classification of focal segmental glomerulosclerosis (FSGS) describes 5 variants, of which the cellular variant is the rarest. This variant is more common in children than in adults, and typically presents with nephrotic-range proteinuria. The cellular variant can be diagnosed after excluding collapsing and tip variants, when there is endocapillary hypercellularity involving at least 25% of the tuft of at least one glomerulus, without immune complexes, and typically with extensive foot process effacement. Prognosis is intermediate between the worst outcomes seen with the collapsing variant versus the best prognosis observed with the tip variant.

Light microscopy: Segmental or global endocapillary hypercellularity, most often focal with usual sclerosis in other glomeruli, foam cells, and overlying hyperplastic visceral epithelial cells.

Immunofluorescence microscopy: No specific staining.

Electron microscopy: Extensive foot process effacement without deposits.

Etiology/Pathogenesis

This cellular variant lesion might be an early abnormality revealed by light microscopy for recurrent FSGS in a kidney transplant. Additional sections may reveal that some of these cellular lesions involve the tubular pole, and thus are better classified as tip lesion variant FSGS.

Differential Diagnosis

The presence of endocapillary hypercellularity revealed by light microscopy suggests possible immune complexes, which can readily be ruled out by lack of deposits shown by immunofluorescence and

electron microscopy. Multiple serial sections are required to assess for possible tip lesions.

Key Diagnostic Features

- Segmental to global endocapillary hypercellularity, typically focal, without immune complex deposits
- Extensive foot process effacement

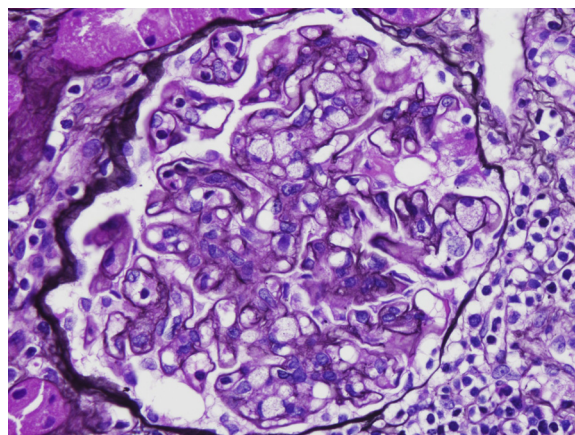


Figure 1. Cellular variant of focal segmental glomerulosclerosis, with endocapillary hypercellularity with foam cells and mononuclear cells (top; Jones silver stain).

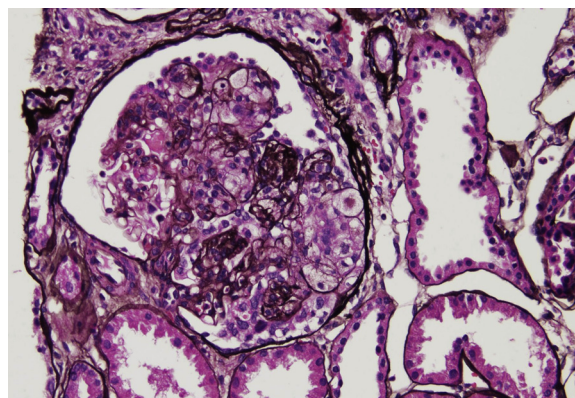


Figure 2. Cellular variant of focal segmental glomerulosclerosis with endocapillary foam cells and mononuclear cells, with overlying visceral epithelial cell hyperplasia (Jones silver stain).

From the ¹Department of Pathology, Microbiology and Immunology, Vanderbilt University, Nashville, TN; and ²Department of Pathology, University of Washington, Seattle, WA.

Support: None.

Financial Disclosure: The authors declare that they have no relevant financial interests.

Address correspondence to agnes.fogo@vanderbilt.edu

Am J Kidney Dis. 66(2):e7.

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0272-6386

<http://dx.doi.org/10.1053/j.ajkd.2015.06.002>